

Calculate SSE.

$$SS_{tot} = SS_t + SS_b + SSE$$

$$678.412 = 426.46 + 110.98 - SSE$$

$$SSE = 140.972$$

calculate Degree of freedom (df)

$$Total\ df = (b \times t) - 1 = (3 \times 8) - 1 = 23$$

$$Treatment\ df = t - 1 = 8 - 1 = 7$$

$$Block\ df = b - 1 = 3 - 1 = 2$$

$$Error\ df = (b - 1)(t - 1) = 2 \times 7 = 14$$

Calculate Mean of Square (MS)

$$MS = \frac{SS}{df}$$

$$MS_t = \frac{SS_t}{df_t}$$

$$= \frac{426.46}{7}$$

$$= 60.922$$

$$MS_b = \frac{SS_b}{df_b}$$

$$= \frac{110.98}{2}$$

$$= 55.49$$

$$MSE = \frac{SSE}{dfe}$$

$$= \frac{140.972}{14}$$

$$= 10.069$$

Calculate F for t, b,

$$F_t = \frac{MSt}{MSe}$$

$$= \frac{60.922}{10.069}$$

$$F_t = 6.050$$

$$F_b = \frac{MSb}{MSe}$$

$$= \frac{55.49}{10.069}$$

$$F_b = 5.510$$

ANOVA Table

Source of variation	SS	df	MS	F	p-value
Treatment	426.46	7	60.922	6.050	< 0.01
Block	110.98	2	55.49	5.510	< 0.05
Error	140.972	14	10.069		
Total	678.412	23			

@ 0.05.

Treatment $F = 2.70$
 $(8, 14) = 2.70$
 $(8, 14) = 2.85$

@ 0.05
 Error $F =$
 $(2, 14)$

$$\frac{2.85}{2} = 1.425$$

to Treatment @ 0.01. $\rightarrow 4.28$. table critical value

$F_{(2,17)} = 4.46$ } mean value 4.3
 $F_{(2,17)} = 4.14$ }

@ 0.05 $\rightarrow$ mean value: 2.77.
 3.81 2.85, 2.70, 2.70 (repeated)

Block

$F_b =$

<u>VI</u>	<u>III</u>	<u>II</u>	<u>I</u>	
4	3	2	1	A
3	2	1	2	B
2	1	2	3	C
		3	1	C

Result.

significance of the tr
 greater

I. since calculated value is less than that of the critical value 4.28 from the table. There is significant effect of treatment (fertilizer dose) in weight of the plant

II. Calculated value is greater than the critical value
 There is significant effect of soil moisture in weight of plant.

The following data is related to the measurement of cholesterol content (%) performed in 4 different laboratories w.r.t 4 diet foods. The design adopted is a RCB D.

Analyse the data and comment on your result.

[Given $F_{0.05} = 3.86$ for (3, 6) D.f]

	I	II	III	IV
A	4	5	6	4
B	3	4	5	8
C	5	3	7	5
D	2	3	4	2

I. Since calculated value is less than table value 4.22, we are not able to reject the null hypothesis. There is no significant effect of diet food on cholesterol content.

II. Calculated value is greater than the critical value of 3.86, therefore there is significant effect of diet food on cholesterol content. To test the effect of diet food on cholesterol content, we follow

Unit - 7

Simple Correlation:

The study of a relationship b/w 2 variables only.

(ie) a dependent and independent variable is known as

Simple Correlation

* For e.g. $y = f(x)$.

$y \rightarrow$ dependent variable

$x \rightarrow$ independent variable

* The correlation coefficient

simple correlation coefficient

$$R_{xy} = \frac{\frac{\sum xy}{n} - \bar{x}\bar{y}}{\left(\sqrt{\frac{\sum x^2}{n} - \bar{x}^2}\right) \left(\sqrt{\frac{\sum y^2}{n} - \bar{y}^2}\right)}$$

Eg: The ice cream sales to tabulate and calculate this.

Date in 2023	Avg daily temp (x)	Daily ice cream sale (y)	x^2	y^2	xy
	14.2	52,750	202	2889062500	763250
2	16.4	81,250	269	6601562500	1332500
3	11.9	46,250	142	2139062500	550375
4	15.2	83,000	231	6889000000	1261600
5	18.5	101,500	342	10302500000	1817750
6	22.1	120,500	488	17030250000	2884050
7	19.4	103000	376	10609000000	1998200
8	25.1	153500	630	23562250000	3852850
9	23.4	136000	548	18496000000	3182400
10	18.1	105250	328	11071562500	1905025

11	22.6	111250	511	1237656500	2514250
12	17.6	102000	296	1040400000	1754400
Total	224.1	1207200	4362	13237562500	2387
Mean	18.7	100604.2	863.5	11031380208.3	1989720.8
Cor. Coeff	0.9575				

$n = 12$

$\bar{x} = 18.675$

$\bar{y} = 100604.2$

$\frac{\sum x^2}{n} = 363.5$

$\frac{\sum y^2}{n} = 11031380208.3$

$\frac{\sum xy}{n} = 1989720.8$

$$R_{xy} = \frac{1989720.8 - (18.675 \times 100604.2)}{\sqrt{(363.5 - 18.675^2)} \sqrt{(11031380208.3 - (100604.2)^2)}}$$

$$= \frac{110937.365}{3.8398 \times 30169.109}$$

$R_{xy} = 0.95764$

Positive and Negative Errors in Correlation.

* Change in magnitude of dependent and independent variables are constant and equal in the same direction is called perfect positive error and vice versa.

* The change in magnitude of the dependent and independent variable are in the same direction for positive errors and vice versa.

Multiple correlation coefficient measures the relationship,

the correlation of multiple correlation between
a host of independent and dependent variables coefficient taken together

$$R_{1,2,3} = \sqrt{\frac{r_{12}^2 + r_{13}^2 - 2r_{12}r_{23}r_{31}}{1 - r_{23}^2}}$$

with $0 \leq R_{1,2,3} \leq 1$.

Note: Symmetry relationship exists with

$$R_{12} = R_{21} ; R_{13} = R_{31} , R_{23} = R_{32} .$$

Partial correlation

$$R_{12,3} = \frac{R_{12} - R_{13}R_{23}}{\sqrt{1 - R_{13}^2} \sqrt{1 - R_{23}^2}}$$

$0 < R_{12,3} \leq 1$.

Any one constant factor. $\rightarrow (n-1)$ factor.

remaining 2 factors are dependent factors.

Note:

* Multiple correlation coefficient is a non-negative

Coefficient (0-1.)

* Coefficient of

* if $R_{1,2,3} = 0 \rightarrow r_{12} = 0, r_{13} = 0$. (multiple determination = $R_{1,2,3}$)

$$R_{1,2,3} \geq r_{12} \text{ (and) } R_{1,2,3} \geq r_{13}$$

* if $R_{1,2,3} = R_{1,32}$

1) From the following data, obtain the multiple correlation coefficients $R_{1.23}$, $R_{2.13}$ and $R_{3.12}$.

X_1	2	5	7	11
X_2	3	6	10	12
X_3	3	6	10	10

Answers: $R_{1.23} = 0.99$, $R_{2.13} = 0.97$ and $R_{3.12} = 0.99$.

$$R_{1.23} = \sqrt{\frac{r_{12}^2 + r_{13}^2 - 2r_{12}r_{23}r_{2313}}{1 - r_{23}^2}} \quad \text{with } 0 \leq R_{1.23}^2 \leq 1$$

$$R_{2.13} = \sqrt{\frac{r_{21}^2 + r_{23}^2 - 2r_{12}r_{13}r_{2313}}{1 - r_{13}^2}}$$

$$R_{3.12} = \sqrt{\frac{r_{31}^2 + r_{32}^2 - 2r_{12}r_{13}r_{2313}}{1 - r_{12}^2}}$$

$$r_{xy} = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sqrt{\sum (x - \bar{x})^2} \sqrt{\sum (y - \bar{y})^2}}$$

$$r_{yz} = \frac{\sum (y - \bar{y})(z - \bar{z})}{\sqrt{\sum (y - \bar{y})^2} \sqrt{\sum (z - \bar{z})^2}}$$

$$r_{zx} = \frac{\sum (z - \bar{z})(x - \bar{x})}{\sqrt{\sum (z - \bar{z})^2} \sqrt{\sum (x - \bar{x})^2}}$$

x_1	x_2	x_3	$(x_1 - \bar{x}_1)$	$(x_2 - \bar{x}_2)$	$(x_3 - \bar{x}_3)$	$(x_1 - \bar{x}_1)^2$	$(x_2 - \bar{x}_2)^2$	$(x_3 - \bar{x}_3)^2$
2	3	1	-4.25	-4.75	-4	18.0625	22.5625	16
5	6	3	-1.25	-1.75	-2	1.5625	3.0625	4
7	10	6	0.75	2.25	1	0.5625	5.0625	1
11	12	10	4.75	4.25	5	22.5625	18.0625	25
			0	-5	0	42.7495	48.75	46

$$\bar{x}_1 = \frac{25}{4} = 6.25$$

$$ESP.O = 828$$

$$\bar{x}_2 = \frac{31}{4} = 7.75$$

$$\bar{x}_3 = \frac{20}{4} = 5$$

$$ESP.O \times ESP.O - ESP.O + ESP.O$$

$$ESP.O - 1$$

$$= 82.19$$

$$r_{12} = \frac{(ESP.O \times ESP.O - ESP.O + ESP.O) + (-4.25 \times -4.75) + (-1.25 \times -1.75) + (0.75 \times 2.25) + (4.75 \times 4.25)}{(42.7495 \times 48.75)}$$

$$\sqrt{(42.7495 \times 48.75)}$$

$$= \frac{44.25}{45.65}$$

$$r_{12} = 0.9693$$

$$r_{13} = \frac{(-4.25 \times -4) + (-1.75 \times -2) + (2.25 \times 1) + (4.25 \times 5)}{(ESP.O \times ESP.O - ESP.O + ESP.O)}$$

$$\sqrt{42.74 \times 46}$$

$$= 51.29$$

$$= \frac{44}{44.34}$$

$$r_{13} = 0.9923$$

$$ESP.O - 1$$

$$r_{28} = \frac{(-4.75 \times -4) + (-1.75 \times -2) + (2.25 \times 1) + (4.25 \times 5)}{46}$$

$$= \frac{18.00 + 3.50 + 2.25 + 21.25}{46} = \frac{44.75}{46} = 0.9713$$

$$r_{23} = 0.9713$$

$$R_{1.28} = \sqrt{\frac{r_{12}^2 + r_{13}^2 - 2r_{12}r_{13}r_{23}}{1 - r_{23}^2}}$$

$$= \sqrt{\frac{0.9693^2 + 0.9923^2 - (2 \times 0.9693 \times 0.9923 \times 0.9713)}{1 - 0.9713^2}}$$

$$= \sqrt{\frac{3.7926}{0.0565} \times \frac{1.9242 - 1.8685}{0.0565}}$$

$$R_{1.23} = 0.9928$$

$$R_{2.13} = \sqrt{\frac{r_{21}^2 + r_{23}^2 - 2r_{12}r_{13}r_{23}}{1 - r_{13}^2}}$$

$$= \sqrt{\frac{0.9698^2 + 0.9713^2 - (2 \times 0.9693 \times 0.9713 \times 0.9713)}{1 - 0.9923^2}}$$

$$= \sqrt{\frac{1.8829 - 1.8685}{0.01534}}$$

$$R_{2.13} = 0.9688$$

$$R_{3.12} = \sqrt{\frac{r_{31}^2 + r_{32}^2 - 2r_{21}r_{13}r_{23}}{1 - r_{12}^2}}$$

$$= \sqrt{\frac{0.9923^2 + 0.9713^2 - (2 \times 0.9693 \times 0.9923 \times 0.9713)}{1 - 0.9693^2}}$$

$$= \sqrt{\frac{1.9280 - 1.8685}{0.0604}}$$

$$R_{3.12} = 0.9925$$

total samples : 10
tappan : 10

$$\frac{10}{4} = 2.5$$

is number of samples

variance : 10
dist food : 10
Experimental Error

total variability = variability in dist + variability in food + variability in experimental error

RCBD Sum.

		lab.				
		I	II	III	IV	
	A	4	5	6	4	$\frac{2808.1 - 1588.1}{4} = 51.0$
Diet	B	3	4	5	8	
	C	5	3	7	5	$\frac{2808.1 - 1588.1}{4} = 51.0$
	D	2	3	4	2	

soln.

Experiment : RCBD

Factors : 2 Laboratories
Diet food.

Total samples : 16

Laboratories : $\frac{\text{samples}}{\text{Diet}}$

$$= \frac{16}{4}$$

= 4 $\rightarrow$ unavoidable variance

Variances : Laboratories
Diet food
Experimental Error

$$\text{Variability}_{\text{tot}} = \text{Variability}_{\text{lab}} + \text{Variability}_{\text{diet}} + \text{Variability}_{\text{Exp. Error}}$$

Diet food	Laboratories				Diet food total Σx_{total}	Diet food Mean $\bar{x}_{total}$
	I	II	III	IV		
A	4	5	6	4	19	4.75
B	3	4	5	8	20	5
C	5	3	7	5	20	5
D	2	3	4	2	11	2.75
Total	14	15	22	19	70	4.3
Mean	3.5	3.75	5.5	4.75	17.5	

$$SS_{total} = SS_L + SS_D + SS_e.$$

$$CT = \frac{(\Sigma x_{total})^2}{n_L \times n_D}$$

$$= \frac{(70)^2}{4 \times 4}$$

$$CT = 306.25$$

$$SS_{total} = \Sigma (x_{total})^2 - CT$$

$$= 348 - 306.25$$

$$SS_{total} = 41.75$$

$$SS_{\text{Total}} = \frac{\sum x_1^2}{n_{D1}} + \frac{\sum x_2^2}{n_{D2}} + \frac{\sum x_3^2}{n_{D3}} + \frac{\sum x_4^2}{n_{D4}} - CT$$

$$= \left(\frac{14^2}{4} + \frac{15^2}{4} + \frac{20^2}{4} + \frac{19^2}{4} \right) - 306.25$$

$$= 316.5 - 306.25$$

$$SS_L = 10.25$$

$$SS_D = \frac{\sum x_1^2}{n_{L1}} + \frac{\sum x_2^2}{n_{L2}} + \frac{\sum x_3^2}{n_{L3}} + \frac{\sum x_4^2}{n_{L4}} - CT$$

$$= \frac{19^2}{4} + \frac{20^2}{4} + \frac{20^2}{4} + \frac{11^2}{4} - 306.25$$

$$= 320.5 - 306.25$$

$$SS_D = 14.25$$

$$SS_{\text{Total}} = SS_L + SS_D + SS_e$$

$$SS_e = 41.75 - (10.25 + 14.25)$$

$$SS_e = 17.25$$

$$\frac{\sum (\text{Inter } x^2)}{n \times n} = TD$$

$$\frac{\sum (07)}{1 \times 1} =$$

$$EC.DOS = TD$$

$$TD - (\text{Inter } x^2) = \text{Inter } x^2$$

$$EC.DOS - 848 =$$

$$31.15 = \text{Inter } x^2$$

calculate degree of freedom (df)

ANOVA table

$$\text{Total df} = n_{\text{tot}} - 1 = (L \times D) - 1 = (4 \times 4) - 1 = 16$$

to check
variation

$$\text{lab df} = 4 - 1 = 3$$

$$\text{Diet df} = 4 - 1 = 3$$

$$\text{Error df} = (L-1)(D-1) = 3 \times 3 = 9$$

calculate Mean of square (MS)

$$MS = \frac{SS}{df}$$

$$MS_L = \frac{SS_L}{df_L}$$

$$MS_D = \frac{SS_D}{df_D}$$

$$MS_e = \frac{SS_e}{df_e}$$

$$= \frac{10.25}{3}$$

$$= \frac{14.25}{3}$$

$$= \frac{17.25}{9}$$

$$MS_L = 3.406$$

$$MS_D = 4.75$$

$$MS_e = 1.916$$

calculate F for lab and Diet food

$$F_L = \frac{MS_L}{MS_e}$$

$$F_D = \frac{MS_D}{MS_e}$$

$$= \frac{3.406}{1.916}$$

$$F_D = \frac{4.75}{1.916}$$

$$F_L = 1.777$$

$$F_D = 2.479$$

ANOVA Table.

(1b) no significant difference

Source of variation	SS	df	MS	F	P
Laboratories	10.25	3	3.406	1.777	0.05
Diet food	14.25	3	4.75	2.479	
Exp. Error	17.25	(21.9)	0.916		
Total	41.75				

Result

- (i) Compare calculated $F_L = 1.777$ with critical value of $F_{(3,9)} = 3.86 \Rightarrow F_L < F_{(3,9)}$

I have concluded that there is no significant

difference in cholesterol % between ~~laboratories~~ laboratories

- (ii) Compare calculated $F_D = 2.479$ with critical value of $F_{(3,9)} = 3.86 \Rightarrow F_D < F_{(3,9)}$

I have inferred that there is no significant difference in cholesterol % between Diet foods.

25.10.2025.

Unit - V

Probability

* It is defined as the chance of occurrence of event.

* success $\rightarrow$ occurrence of event indicated as P .

failure $\rightarrow$ Probability is indicated as $(1-P) = Q$.

H	H	H	H
T	H	T	H
H	T	H	T

Pantheerselvam book Pg.No: 169.

Sample space : It is the list of all possible outcomes of Experiment.

Event : The occurrence of a particular outcome or a combination of outcomes is called an event of the experiment.

Exp: Tossing of 3 coins at a time.

sample space = 8.

C1	C2	C3	
H	H	H	①
H	H	T	②
H	T	H	③
H	T	T	④
T	H	H	⑤
T	H	T	⑥
T	T	H	⑦
T	T	T	⑧

Events = 4.

H H H 1

T T T 1

H H T 3

H T T 3

Frequency = 1, 3, 1.

*

Probability = $\frac{\text{No. of occurrence of Event}}{\text{Total no. of Outcomes}}$

Probability

Eg: Tossing 2 coins at a time.

Outcomes

H	H	HH
H	T	HT
T	H	TH
T	T	TT

Probability of occurrence of HT = $\frac{2}{4} = 0.5$.

Probability of occurrence of HH = $\frac{1}{4} = 0.25$

TT = $\frac{1}{4} = 0.25$

Probability distribution

Types:

1. Discrete probability distribution
2. Continuous probability distribution.

1. Discrete Probability distribution

* The outcomes can be count is discrete PD.
Eg: Tossing coins.

①	H	H	H
②	T	H	H
③	H	T	H
④	H	H	T
⑤	T	T	T
⑥	H	T	T
⑦	T	T	T

Ex: Tossing of 3 coins at a time.

$X \rightarrow$ random variable is defined as a discrete random variable indicating the number of event of head occurrence.

The events = 4 (HHH, TTT, HHT, HTT).

$X=0 \rightarrow$ Head occurrence = 0 $\rightarrow$ TTT.

$X=1 \rightarrow$ " " = 1 $\rightarrow$ HTT

$X=2 \rightarrow$ " " = 2 $\rightarrow$ HHT

$X=3 \rightarrow$ " " = 3 $\rightarrow$ HHH

Based on the assumption, the probability distribution is.

$$P(X) = \begin{cases} 1/8 & \text{if } X=0 \\ 3/8 & \text{if } X=1 \\ 3/8 & \text{if } X=2 \\ 1/8 & \text{if } X=3 \end{cases}$$

or otherwise.

This is called Probability Mass Function (P.M.F).

$$\sum_{i=1}^n P_i(X) = 1 \rightarrow \text{cumulative distribution function.}$$

where $n = \text{no. of events} = 4$.

event

Mean of Binomial distribution = np

2. Continuous Probability Distribution

x has a range (Random variable).

Eg: $f(x) = 3x^2 - 3x + 1.5$ where $0 \leq x \leq 1$

otherwise.

for a probability distribution the value on the cumulative distribution is specified range should be

4.

The value of the C.D can be obtained by.

$$\int_0^1 f(x) dx = \int_0^1 (3x^2 - 3x + 1.5) dx = \left[\frac{3x^3}{3} - \frac{3x^2}{2} + 1.5x \right]_0^1$$

= 1.

Binomial Probability Distribution:

A random variable x is said to follow Binomial distribution if it assumes non-negative values and its

probability mass function is given by

$$P(X=x) = p(x) = {}^n C_x p^x q^{n-x}, \quad x=0,1,2,\dots$$

$$q=1-p.$$

n = no. of trials.

where

p = denotes probability of success

q = probability of failure

np = Mean of Binomial distribution

npq = variance of binomial distribution.

Properties

- (1) There are only two outcomes namely success and failure, and these are denoted by p and q such that $p+q=1$.
- (2) Each trial is independent of others
- (3) The number of trials is finite, namely ' n '.

Eg :

x	f
0	50
1	20
2	15
3	10
4	3
5	2

x : No. of seeds germinated

f : No. of pots.

$$n=5$$

$$x=0, 1, 2, 3, 4, 5$$

$$p(x) = {}^5C_x p^x q^{5-x}$$

Applications

* Statistical Quality Control

* Plant Studies

* Military Applications (Hit/Miss).

Note :

$$n(x) = \frac{n!}{x!(n-x)!}$$

Poisson Distribution

A random variable x is said to follow Poisson distribution if it assumes non-negative values and

its probability mass function is given by

$$P(X=x) = \frac{e^{-\lambda} \lambda^x}{x!}, \quad \begin{cases} x = 0, 1, 2, \dots \\ \lambda > 0 \end{cases}$$

0, otherwise

λ = mean

λ = variance

Properties

(1) It is the limiting case of Binomial distribution

Under the following conditions, binomial distribution may lead to Poisson distribution.

(i) Let n be the no. of trials, $n \rightarrow \infty$

(ii) Let the probability of success be $p \rightarrow p \rightarrow 0$

(iii) np is finite

(2) Poisson distribution can be used for rare events such as deaths, mistakes etc.

Eg: Distribution of occurrence of flood in 1900-2000

$$\frac{1}{(x-1)!} = \frac{1}{(x-1)!}$$

Application

1) Quality control

2) Quality Assurance

Problems : Binomial distribution

- 1) If a coin is tossed for 10 times, what is the probability of having (a) 0 heads; (b) 3 heads (c) at most 2 heads (d) at least 3 heads?

Given :

no. of trials $n = 10$

To find :

- a) 0 head $P = ?$
b) 3 " $P = ?$
c) at most 2 heads $P = ?$
d) at least 3 heads $P = ?$

Soln :

a) 0 head

Poisson distribution is applied, p for occurrence of head $= \frac{1}{2} = 0.5$.

$$\therefore P(x) = {}^n C_x p^x q^{n-x}$$

$$P(x=0) = {}^{10} C_0 (0.5)^0 (0.5)^{10-0} \\ = 0! \times 1 \times (0.5)^5$$

$$P(x=0) = 0.0009765$$

b) 8 heads:

$$P(X=3) = {}^{10}C_3 (0.5)^3 (0.5)^{10-3}$$

$$= \frac{10!}{3! (10-3)!} \times 0.125 \times 0.007812$$

$$= \frac{10 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}{3! \times 7!} \times 0.0001953$$

$$= \frac{10 \times 9 \times 8}{3 \times 2 \times 1} \times 0.0001953$$

$$= 120 \times 0.0001953$$

$$P(X=3) = 0.11719$$

c) at most 2 heads

at most = maximum

$$P(X \leq 2) = \sum_{x=0}^2 n C_x p^x q^{n-x}$$

$$= {}^{10}C_0 (0.5)^0 (0.5)^{10}$$

... ..

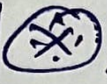
$$P(X) = {}^{10}C_x p^x q^{10-x}$$

$$P(X=0) = {}^{10}C_0 (0.5)^0 (0.5)^{10}$$

$$= 1 \times 1 \times 0.0001953$$

$$P(X=0) = 0.0001953$$

d) at least 3 heads

at least = Minimum 

$$P(X \geq 3) = \sum_{x=3}^{10} {}^{10}C_x p^x q^{n-x}$$

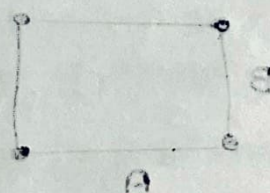
$$= 1 - \sum_{x=0}^2 {}^{10}C_x p^x q^{n-x}$$

$$= 1 - \left[{}^{10}C_0 (0.5)^0 (0.5)^{10} + {}^{10}C_1 (0.5)^1 (0.5)^9 + {}^{10}C_2 (0.5)^2 (0.5)^8 \right]$$

$$= 1 - \left[(1 \times 1 \times 0.5^{10}) + \left(\frac{10!}{1!9!} (0.5) \times (0.5)^9 \right) + \left(\frac{10!}{2!8!} (0.5)^2 (0.5)^8 \right) \right]$$

1 - $\frac{1}{2^{10}}$  $\frac{1}{2}$

A B
 1 1
 1 1
 1 1
 1 1
 1 1



(iii) combined AB
 (i) alone A
 (ii) alone B

